

Finding the equilibrium endpoints of a fair Rock–Paper–Scissors game

This note walks through how the symmetric Nash equilibria of a “fair” generalized Rock–Paper–Scissors game are computed, and in particular how to find the *endpoints* (extreme points / vertices) of the equilibrium set. The running example is the $n = 4$ “cop” game, whose equilibrium set is a line segment with two endpoints.

1 Setup

A game on n moves is encoded by an $n \times n$ *antisymmetric* sign matrix $A = M$ with $A = -A^\top$ and entries $A_{ij} \in \{-1, 0, +1\}$: row i beats column j when $A_{ij} = +1$, loses when $A_{ij} = -1$, ties when $A_{ij} = 0$. Because the game is symmetric and zero-sum, its value is 0 and

$$p^\top A p = 0 \quad \text{for every } p.$$

A mixed strategy is a point of the probability simplex

$$\Delta = \{ p \in \mathbb{R}^n : p_i \geq 0, \sum_i p_i = 1 \}.$$

A strategy p is a (symmetric) Nash equilibrium iff it is a best response to itself, i.e. no pure move earns a positive payoff against it:

$$(Ap)_i \leq p^\top A p = 0 \quad \text{for all } i.$$

Hence the equilibrium set is the polytope

$$O = \{ p \in \Delta : Ap \leq \mathbf{0} \}. \tag{1}$$

O is convex, so the full set of equilibria is the convex hull of its *vertices*. “How many solutions” a game has is the number of those vertices: 1 means a unique equilibrium, 2 means a segment, more means a polygon or higher-dimensional polytope.

2 Fully-mixed equilibria live in the kernel

If p is *fully mixed* ($p_i > 0$ for all i), then complementary slackness forces equality everywhere: since

$$0 = p^\top A p = \sum_i p_i (Ap)_i,$$

with every $p_i > 0$ and every $(Ap)_i \leq 0$, each term must vanish, so $(Ap)_i = 0$ for all i . Therefore

$$\text{a fully-mixed equilibrium} \implies Ap = \mathbf{0}, \quad \text{i.e.} \quad p \in \ker(A).$$

The fully-mixed equilibria are exactly the strictly positive vectors of $\ker(A)$, rescaled to sum to 1.

Theorem 1 (Parity). *A real antisymmetric matrix has even rank. Consequently $\dim \ker(A) = n - \text{rank}(A)$ has the same parity as n , and a fully-mixed equilibrium can be unique (i.e. $\dim \ker(A) = 1$) only when n is odd. For even n a fully-mixed equilibrium is never isolated: it sits in a family of dimension $\dim \ker(A) - 1 \geq 1$.*

Why the rank is even. The eigenvalues of a real antisymmetric matrix are purely imaginary and occur in conjugate pairs $\pm i\lambda$; equivalently A is orthogonally similar to a block-diagonal matrix of 2×2 blocks $\begin{pmatrix} 0 & \lambda \\ -\lambda & 0 \end{pmatrix}$ plus a zero block. Each nonzero block contributes 2 to the rank, so $\text{rank}(A)$ is even. $\square$

3 Worked example: the cop game

Label the moves Cop = 0, K9 = 1, Perp = 2, Witness = 3, with

$$\text{Witness} > \text{Cop}, \quad \text{Cop} > \text{Perp}, \quad \text{Cop} > \text{K9}, \quad \text{K9} > \text{Perp}, \quad \text{Perp} > \text{Witness},$$

and the single tie K9 vs. Witness. With $A_{ij} = +1$ when i beats j ,

$$A = \begin{pmatrix} 0 & 1 & 1 & -1 \\ -1 & 0 & 1 & 0 \\ -1 & -1 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix}.$$

3.1 Step 1 — write the homogeneous system

The fully-mixed equilibria satisfy $Ap = \mathbf{0}$. Multiplying A by the unknown vector $p = (p_0, p_1, p_2, p_3)^\top$ and reading off one equation per row:

$$\begin{aligned} (\text{row 0, Cop}) : & \quad 0p_0 + 1p_1 + 1p_2 - 1p_3 = 0, \\ (\text{row 1, K9}) : & \quad -1p_0 + 0p_1 + 1p_2 + 0p_3 = 0, \\ (\text{row 2, Perp}) : & \quad -1p_0 - 1p_1 + 0p_2 + 1p_3 = 0, \\ (\text{row 3, Witness}) : & \quad 1p_0 + 0p_1 - 1p_2 + 0p_3 = 0. \end{aligned}$$

3.2 Step 2 — row reduce A to find the kernel

Work on A directly (the right-hand side stays $\mathbf{0}$ throughout). Write the rows as $R_0, \dots, R_3$:

$$\begin{bmatrix} 0 & 1 & 1 & -1 \\ -1 & 0 & 1 & 0 \\ -1 & -1 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \xrightarrow{R_0 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & -1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & -1 & 0 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}.$$

Now clear column 0 below the pivot in row 0 with $R_1 \leftarrow R_1 + R_0$ and $R_2 \leftarrow R_2 + R_0$:

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Finally clear column 1 with $R_2 \leftarrow R_2 + R_1$:

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \text{RREF}(A).$$

There are two nonzero rows, so $\text{rank}(A) = 2$ and

$$\dim \ker(A) = n - \text{rank}(A) = 4 - 2 = 2,$$

consistent with Theorem 1 ($n = 4$ even $\Rightarrow$ rank even).

3.3 Step 3 — read off the kernel basis

The two nonzero rows give the relations

$$p_0 - p_2 = 0, \quad p_1 + p_2 - p_3 = 0.$$

Since the rank is 2, two of the four unknowns are free. We choose p_1 and p_2 as the free parameters (this is the choice that makes the final answer cleanest) and solve the two relations for the remaining two:

$$p_0 = p_2, \quad p_3 = p_1 + p_2.$$

Set each free variable to 1 in turn (the other to 0) to get a basis of $\ker(A)$:

$$\begin{aligned} p_1 = 1, p_2 = 0 : \quad p_0 = 0, p_3 = 1 &\Rightarrow v_1 = (0, 1, 0, 1), \\ p_1 = 0, p_2 = 1 : \quad p_0 = 1, p_3 = 1 &\Rightarrow v_2 = (1, 0, 1, 1). \end{aligned}$$

A quick check: $Av_1 = \mathbf{0}$ and $Av_2 = \mathbf{0}$ (e.g. row 0 on v_2 gives $0 \cdot 1 + 1 \cdot 0 + 1 \cdot 1 - 1 \cdot 1 = 0$). Writing $b := p_1$ and $a := p_2$, every kernel vector is

$$p = b v_1 + a v_2 = (a, b, a, a + b). \quad (2)$$

3.4 Step 4 — impose the simplex

A strategy must satisfy $\mathbf{1}^\top p = 1$ and $p \geq \mathbf{0}$. Summing the components of (2),

$$p_0 + p_1 + p_2 + p_3 = a + b + a + (a + b) = 3a + 2b,$$

so the normalization condition is

$$3a + 2b = 1, \quad a \geq 0, b \geq 0. \quad (3)$$

The single linear equation $3a + 2b = 1$ cuts the 2-dimensional kernel down to a 1-dimensional line in the (a, b) -plane; the inequalities $a \geq 0, b \geq 0$ clip that line to a bounded **segment**. The equilibrium set is therefore a 1-dimensional family, matching $\dim \ker(A) - 1 = 2 - 1 = 1$.

4 Finding the endpoints

A point of the segment (3) is an *endpoint* (vertex) exactly when one of the two non-negativity constraints becomes tight — i.e. $a = 0$ or $b = 0$. Setting each to equality in turn and solving the remaining 1-variable equation:

Endpoint 1: $b = 0$. The constraint $3a + 2b = 1$ becomes $3a = 1$, so

$$a = \frac{1}{3}, \quad b = 0.$$

Substituting into (2) ($p = (a, b, a, a + b)$):

$$p^{(1)} = \left(\frac{1}{3}, 0, \frac{1}{3}, \frac{1}{3}\right).$$

This drops K9 ($p_1 = 0$).

Endpoint 2: $a = 0$. Now $3a + 2b = 1$ becomes $2b = 1$, so

$$a = 0, \quad b = \frac{1}{2}.$$

Substituting into (2):

$$p^{(2)} = \left(0, \frac{1}{2}, 0, \frac{1}{2}\right).$$

This drops Cop and Perp ($p_0 = p_2 = 0$).

4.1 Verify each endpoint is an equilibrium

Each endpoint must satisfy the equilibrium condition $Ap \leq \mathbf{0}$. Because both lie in $\ker(A)$ by construction, we should in fact get $Ap = \mathbf{0}$ exactly. Check $p^{(1)} = (\frac{1}{3}, 0, \frac{1}{3}, \frac{1}{3})$ row by row:

$$Ap^{(1)} = \begin{pmatrix} 0(\frac{1}{3}) + 1(0) + 1(\frac{1}{3}) - 1(\frac{1}{3}) \\ -1(\frac{1}{3}) + 0(0) + 1(\frac{1}{3}) + 0(\frac{1}{3}) \\ -1(\frac{1}{3}) - 1(0) + 0(\frac{1}{3}) + 1(\frac{1}{3}) \\ 1(\frac{1}{3}) + 0(0) - 1(\frac{1}{3}) + 0(\frac{1}{3}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \leq \mathbf{0} \checkmark$$

Similarly for $p^{(2)} = (0, \frac{1}{2}, 0, \frac{1}{2})$:

$$Ap^{(2)} = \begin{pmatrix} 1(\frac{1}{2}) - 1(\frac{1}{2}) \\ 0 \\ -1(\frac{1}{2}) + 1(\frac{1}{2}) \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \leq \mathbf{0} \checkmark$$

Every fully-mixed (strictly interior) equilibrium is a convex combination $\lambda p^{(1)} + (1 - \lambda) p^{(2)}$ with $0 < \lambda < 1$ — the open interior of the segment. The endpoints themselves are boundary equilibria, each leaving out at least one move.

4.2 Picking the canonical point: leximin, step by step

To report a single set of odds we take the leximin (max–min) point: the equilibrium that maximizes the smallest play rate. On the segment the four coordinates are $(a, b, a, a + b)$. Since $a, b \geq 0$ we have $a + b \geq a$ and $a + b \geq b$, so the smallest coordinate is

$$\min_i p_i = \min(a, b).$$

We maximize $m := \min(a, b)$ subject to $3a + 2b = 1$.

Upper bound. Any feasible point has $a \geq m$ and $b \geq m$, hence

$$1 = 3a + 2b \geq 3m + 2m = 5m \implies m \leq \frac{1}{5}.$$

Attainability. The bound is met by balancing the two free variables, $a = b$:

$$3a + 2a = 5a = 1 \implies a = b = \frac{1}{5},$$

which gives $\min(a, b) = \frac{1}{5}$. Substituting $a = b = \frac{1}{5}$ into (2):

$$p^* = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{2}{5}\right) = (0.2, 0.2, 0.2, 0.4).$$

This is the midpoint-in-spirit of the segment — the equilibrium farthest from every boundary face — guaranteeing each move at least a 20% play rate.

Proposition 1 (Endpoints always touch the boundary). *If the equilibrium family has positive dimension, every one of its vertices lies on a face $\{p_i = 0\}$ of the simplex. Equivalently, a vertex of O can be strictly interior only when $\dim \ker(A) = 1$ (the unique-equilibrium, odd- n case).*

Sketch. Within the hyperplane $\mathbf{1}^\top p = 1$, a vertex is a point where $n - 1$ of the facet inequalities $\{p_i \geq 0\} \cup \{(Ap)_i \leq 0\}$ are active with linearly independent normals. An interior point ($p_i > 0$ for all i) activates none of the $p_i \geq 0$ facets, so all $n - 1$ active facets must come from $(Ap)_i = 0$. But an interior equilibrium already has $Ap = \mathbf{0}$, and for the active rows of A together with $\mathbf{1}$ to span $\mathbb{R}^n$ one needs $\text{rank}(A) = n - 1$, i.e. $\dim \ker(A) = 1$. Hence when $\dim \ker(A) \geq 2$ no vertex is interior. $\square$ $\square$

5 The general recipe

For an arbitrary fair game A , the same procedure finds all extreme equilibria:

1. **Kernel.** Solve $Ap = \mathbf{0}$ (compute $\ker A$ via its null space) to parametrize the candidate fully-mixed equilibria.
2. **Normalize and clip.** Intersect $\ker A$ with the simplex $\{\mathbf{1}^\top p = 1, p \geq \mathbf{0}\}$.
3. **Enumerate vertices.** A vertex of O in (1) is the unique solution of the $n \times n$ system formed by $\mathbf{1}^\top p = 1$ together with $n - 1$ tight facets chosen from $\{p_i = 0\} \cup \{(Ap)_i = 0\}$ whose normals are linearly independent; keep each solution that is feasible ($p \geq \mathbf{0}, Ap \leq \mathbf{0}$), and deduplicate. The number of distinct vertices is the number of solutions of the game.

For the cop game this yields exactly the two endpoints above. More symmetric games (RPS, RPSLS, the Paley tournament Q_7) have $\dim \ker(A) = 1$ and a single vertex — the unique uniform equilibrium — while larger even- n games can have equilibrium *polygons* with several vertices.

Canonical representative. When the equilibrium set is not a single point, a canonical representative is needed to report “the odds.” We use the *leximin* (max–min) point: maximize the smallest play rate, then the next, and so on. For the cop game this is the centre $(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{2}{5}) = (0.2, 0.2, 0.2, 0.4)$ — the point of the segment farthest from every boundary face, guaranteeing each move at least a 20% play rate.

6 Modular decomposition: how the equilibrium count factorizes

Many games are not “new”: they are a smaller game with some moves replaced by whole sub-games. Modular decomposition makes this precise and shows that the number of equilibria multiplies over the construction.

6.1 Modules and substitution

A *module* is a set of moves $S \subseteq V$ that every outside move relates to identically: for each $x \notin S$ the sign $A_{x,a}$ is the same for all $a \in S$ (so x beats all of S , loses to all of S , or ties all of S). A tie-twin pair is exactly a size-2 all-tie module. A game with no module other than singletons and V is **modular-prime**; otherwise it is a *substitution*

$$G = H[M_1, \dots, M_k],$$

where the *quotient* H is a $k \times k$ game on the modules and move i of H is blown up into the sub-game M_i . The relation between two modules is well defined (a single sign), so H is itself a skew $\{-1, 0, +1\}$ game.

6.2 The factorization law

Write a strategy of G as a mass w_i on each module i together with a conditional distribution q_i inside it. For a move $a \in M_j$,

$$(A_G p)_a = \underbrace{\sum_{i \neq j} h_{ji} w_i}_{E_j \text{ (same for all } a \in M_j)} + w_j (A_{M_j} q_j)_a,$$

because every move of M_j sees the other modules identically (that is the module property), and $h_{ji} = (A_H)_{ji}$ is the quotient sign. Summing the equilibrium conditions over the support of module j , weighted by q_j , and using $q_j^\top A_{M_j} q_j = 0$, gives $E_j = 0$ on every played module. Hence

p is an equilibrium of $G \iff w$ is an equilibrium of H , and each played q_j is an equilibrium of M_j .

So the equilibrium polytope is a product, $O(G) = O(H) \times \prod_j O(M_j)$, and its vertices are products of vertices:

$$n_{eq}(G) = \sum_{\text{vertices } w \text{ of } O(H)} \prod_{j \in \text{supp}(w)} n_{eq}(M_j). \quad (4)$$

When the quotient has a *unique, fully supported* equilibrium (e.g. RPS, RPSLS), (??) collapses to the clean product $n_{eq}(G) = \prod_j n_{eq}(M_j)$.

6.3 Reading off the special cases

- **All-tie quotient** ($H = \mathbf{0}$, the k modules mutually tie): $O(H)$ is the whole simplex, whose vertices are the k unit vectors, so $n_{eq} = \sum_j n_{eq}(M_j)$. A tie-twin pair is the case $k = 2$ with single moves: $n_{eq} = 1 + 1 = 2$.
- **Transitive (“order”) quotient**: a dominant module is played purely, so n_{eq} equals that module’s count.
- **Prime quotient**: the general mixed sum (??).

The decomposition is recursive — modules nest — so n_{eq} is computed bottom-up over the decomposition tree.

6.4 Worked example: the $n = 6$ game with eight equilibria

Take RPS and replace each of its three moves by a tie-twin pair: $G = \text{RPS}[\text{tie}_2, \text{tie}_2, \text{tie}_2]$ on $n = 6$ moves. The quotient is RPS, which has the unique fully-supported equilibrium $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, and each module is an all-tie pair with $n_{eq}(\text{tie}_2) = 2$. By (??),

$$n_{eq}(G) = n_{eq}(\text{RPS}) \cdot 2 \cdot 2 \cdot 2 = 1 \cdot 8 = 8.$$

Each vertex chooses, for every twin pair, which copy carries that pair's $\frac{1}{3}$ mass — the $2^3 = 8$ corners. The leximin point instead splits each pair evenly, giving the uniform $\frac{1}{6}$ on all six moves (the centre of the polytope).

6.5 Why this is the right notion of “irreducible”

The blow-up in (??) is purely a substitution effect, so the games with many equilibria are exactly the heavily-substituted ones. Twin-free only forbids all-tie size-2 modules; it still admits substituting a prime sub-game with $n_{eq} \geq 2$ (e.g. a cop module), which is why $\text{RPSLS}[\text{cop}^{\times 5}]$ is twin-free yet has $n_{eq} = 2^5 = 32 > 20 = n$. Modular-prime games — no module at all — are the genuine irreducible atoms; empirically every prime fair game enumerated satisfies $n_{eq} \leq n$, while the twin-free bound is false. The ring games C_n (each move beats the next, ties the rest) are prime for all $n \geq 3$: any proper subset has a boundary move that an outside neighbour beats while it ties the interior, so no nontrivial module exists.